MILITARY SPECIFICATION

ARMOR, STEEL, PLATE, WROUGHT, (ESR) (3/16 THROUGH 3 INCHES, INCLUSIVE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers electro-slag remelted (ESR), wrought-steel homogeneous armor plate in thicknesses from 3/16 through 3 inches inclusive (see 6.1).

1.2 Classification. Wrought armor shall be of the following classes, as specified.

1.2.1 Class 1. Wrought armor plate which is heat treated to develop maximum resistance to penetration.

1.2.2 Class 2. Wrought armor plate which is heat treated to develop maximum resistance to shock.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

FEDERAL
Fed. Test Method Std. No. 151 - Metals; Test Methods

MILITARY
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 9515

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

E 8 - Tension Testing of Metallic Materials
E 10 - Brinell Hardness of Metallic Materials
E 18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials
E 23 - Notched Bar Impact Testing of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2) and before production has commenced, a sample or samples of the specified item shall be made available to the contracting officer or his authorized representative for approval in accordance with 4.2.1. The approval of the first article samples authorizes the commencement of production but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The first article samples and test plates shall be manufactured by the process proposed for use on production armor.

3.2 Acceptance requirements.

3.2.1 Materials.

3.2.1.1 Structure. The process of manufacture shall be such as to produce armor having, as nearly as practicable, a homogeneous structure throughout.

3.2.2 Chemical composition. The chemical (check) analysis of the ingots shall conform to the requirements of table I.

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Table I. Chemical composition, percent

Elemement	
Carbon	0.31 - 0.35
Manganese	0.60 - 0.90
Silicon	0.20 - 0.30
Phosphorous	0.005 max.
Sulphur	0.005 max.
Nickel	2.75 min.
Chromium	0.65 - 0.90
Molybdenum	0.30 - 0.50
Oxygen	25 ppm max.

3.2.3 Heat treatment. All plates (class 1) in each lot, including samples, shall receive the same heat treatment except for such variations in tempering temperature as may be necessary to produce the prescribed hardness. In no case should the tempering temperature exceed 425°F for class I armor. The austenitizing temperature may vary within a range of 50° above the temperature used for test plates, but in no case shall it exceed 1550°F.

3.2.4 Processing controls.

3.2.4.1 Heating. Local or general heating shall not be performed after the final quenching and tempering operation, except as provided elsewhere in this specification. A detailed outline of the procedure to be used in each operation of the following processes shall be submitted in writing for authorization by the procuring activity.

3.2.4.1.1 Edge preparation. Oxygen cutting or beveling of edges shall be permitted after final heat treatment provided the procedure is such that no cracks develop on any oxygen-cut edge.

3.2.4.1.2 Repairing. Unless otherwise approved by the procuring activity, weld repairs shall not be permitted.

3.2.4.1.3 Forming. Forming after the final quenching and tempering operations shall not be done except when authorized by the procuring activity.

3.2.4.2 Surface imperfections. Surface imperfections may be removed by grinding providing the serviceability of the material is not impaired and the grinding does not result in the reduction of the plate thickness in the ground area to less than the tolerances specified in tables II or III as applicable.

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3.2.4.2.1 Grinding. In removing imperfections, the total ground area shall not exceed 24 square inches in any square foot of plate surface, and the minimum distance between ground areas shall be 2-1/2 inches.

3.2.4.2.2 Determination of minimum plate thickness. The minimum plate thickness shall be determined by subtracting the depth of the removed defect, measured by depth or feeler gage, from the actual gage of the plate. When removed defects occur on both sides of a plate and overlap in any location the sum of the two depths shall be deducted from the actual gage of the plate.

3.2.5 Mechanical properties.

3.2.5.1 Hardness. The average surface hardness of each plate including samples shall be within the range shown in table IV for the applicable class and thickness. The diameters of Brinell hardness impressions determined on the surface of any plate or sample shall not vary by more than 0.15 mm respectively, between the maximum and minimum values.

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Table II. Thickness tolerances for plates up to 2" thick

Thickness tolerances in inches over and under ordered thickness[1/] for a single plate 2 inches and under in thickness[2/]

Specified Thickness Inches	To 60"	Tolerances over ordered thickness for widths given: 60" to 72" excl.	72" to 84" excl.	84" to 96" excl.	96" to 108" excl.	108" to 120" excl.	120" to 132" excl.	132" to 144" excl.	144" to 168" excl.	168" to 182" excl.	182" and over
Up to 1/4	2/	.016	.016	.019	.019	.023	----	----	----	----	----
5/16		.016	.019	.019	.019	.023	.026	----	----	----	----
3/8		.016	.019	.019	.023	.023	.026	----	----	----	----
7/16		.016	.019	.019	.023	.026	.026	.031	----	----	----
1/2		.016	.019	.019	.023	.026	.026	.031	.036	----	----
9/16		.019	.019	.019	.023	.026	.031	.031	.036	----	----
5/8		.019	.019	.019	.023	.026	.031	.031	.036	----	----
11/16		.019	.019	.019	.023	.026	.031	.031	.036	----	----
3/4		.019	.019	.023	.023	.026	.031	.039	.043	----	----
13/16		.023	.023	.023	.026	.031	.031	.039	.043	----	----
7/8		.023	.023	.026	.026	.031	.036	.039	.043	----	----
15/16		.023	.023	.026	.026	.031	.036	.043	.048	----	----
1		.026	.026	.026	.026	.031	.036	.043	.048	----	----
1-1/16		.026	.026	.026	.031	.031	.036	.043	.048	----	----
1-1/8		.026	.026	.026	.031	.031	.039	.043	.048	----	----
1-3/16		.031	.031	.031	.031	.036	.043	.048	.053	----	----
1-1/4		.031	.031	.031	.036	.036	.043	.048	.053	----	----
1-5/16		.031	.031	.031	.036	.036	.043	.053	.058	----	----
1-3/8		.031	.031	.031	.036	.039	.048	.053	.058	----	----
1-7/16		.036	.036	.036	.036	.043	.048	.058	.063	----	----
1-1/2		.036	.036	.036	.039	.043	.048	.058	.063	----	----
1-9/16		.036	.036	.036	.039	.043	.053	.058	.070	----	----
1-5/8		.036	.036	.036	.043	.048	.053	.063	.070	----	----
1-11/16		.039	.039	.039	.043	.048	.058	.063	.070	----	----
1-3/4		.039	.039	.039	.043	.048	.058	.068	.078	----	----
1-13/16		.043	.043	.043	.048	.053	.058	.068	.078	----	----
1-7/8		.043	.043	.043	.048	.053	.063	.068	.078	----	----
1-15/16		.043	.043	.043	.048	.053	.063	.076	.084	----	----
2		.043	.043	.043	.048	.053	.063	.076	.084	----	----

1/ For intermediate thickness, the tolerance of the closer specified gage shall apply. In case of mid-point, the tolerance for lower gage or interpolated value shall apply.

2/ Plates under 60" are rolled double width and the equivalent wider plate tolerances shall apply.

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Table III. Thickness tolerances for plates over 2" thick

Thickness tolerances in inches over and under ordered thickness for single plate over 2 inches in thickness when ordered to thickness in inches.

Tolerance over and under specified thickness for widths given:

Specified thickness inches	Tolerances over and under specified thickness for widths given				
	Under 60" excl.	60" to 84" excl.	84" to 120" excl.	120" to 132" excl.	132" and over
2-3	1/	.060	.068	.068	.076

1/ Plates ordered under 60" are rolled double width and the equivalent wider plate tolerance shall apply.

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Table IV. Brinell hardness requirements

Specified nominal thickness of plate, inches	Armor class	HB range (3000 Kg. load)
From 3/16 up to 1	1	534 - 601
From 1 to 3 incl.	1	477 - 534
From 3/16 to 3 incl.	2	302 - 352

3.2.5.2 Impact resistance. The Charpy V-notch impact resistance of armor submitted shall meet the requirements shown in table V for the applicable hardness and thickness. The Charpy V-notch impact test specimens shall be obtained in the TL orientation, i.e., the specimens shall be taken transverse to the major direction of rolling with the notch running in the longitudinal direction (i.e., the crack will propagate in the longitudinal direction).

Table V. Minimum Charpy V-notch impact resistance requirements (at -40°F ± 2°F[1/])

Brinell hardness (HB) range, average	Minimum impact resistance (average of the two or more test specimens) ft. lbs.
302 - 352	[2/]
477 - 534	23
534 - 601	20

[1/] TL specimen orientation.

[2/] The impact resistance requirement for class 2 armor shall be as agreed upon between the procuring agency and the supplier.

3.2.6 Ballistic resistance (class-1 armor only).

3.2.6.1 Resistance to penetration at normal obliquity. The minimum requirements for acceptable ballistic limits shall be as specified in table VI.

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3.2.6.2 Resistance to spalling.

3.2.6.2.1 Plates 1/2 to 11/16 inch in thickness. Spalling shall not exceed 1-5/8 inches when tested with 20 mm AP M95 ammunition at 0° obliquity in the velocity range of 2500 - 2550 fps. Two rounds are to be fired at the specified velocity level.

3.2.6.2.2 Plates 3/4 inch and greater in thickness. Spalling shall not exceed 2.0 calibers in diameter after proof firing.

3.2.6.3 Cracking.

3.2.6.3.1 Plates 1/4 inch through 2 inches in thickness. Plates shall not within 24 hours after ballistic testing, develop any crack in the front or the rear of the plate that extends outside or is entirely outside a circle having a radius of 2.0 calibers (or a radius of 1 inch for the caliber .30 AP test) measured from the center of the impact of the projectile used.

3.2.7 Dimensions and tolerances.

3.2.7.1 Dimensions. Plates shall comply with the dimensions specified on the applicable drawings or in the contract or order (see 6.2).

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Table VI. Minimum V_{50} protection ballistic limit at 0° obliquity

Thickness (in.)	Projectile	(BL)
3/16	Cal .30 AP M2	1700
1/4		2000
5/16		2300
3/8		2500
7/16		2900
1/2	Cal .50 AP M2	2250
5/8		2500
3/4		2700
7/8		3000
1/2	14.5 mm AP B32	1800
3/4		2600
1		3000
1	14.5 API BS41	2400
1-1/4		3000
1-1/2		3400
1	20 mm HVAP-T M602	1700
1-1/4		2100
1-1/2		2500
2		3000

3.2.7.2 Thickness. The thickness tolerance of each plate, after final heat treatment, shall be in accordance with table II or table III for the thickness specified.

3.2.7.3 Flatness. The flatness tolerance of each plate shall be within 1/2 inch from a straight line 6 feet in length. Tighter requirements may be specified in the contract or order and shall be as agreed upon between the contractor and the procuring activity.

3.2.8 Surface decarburization. For plates heat treated to HB534 and above, the depth of decarburization after all heat treat operations shall not exceed 0.003 inch on any surface. For plates heat treated to less than HB534, depth of decarburization after all heat treat operations shall not exceed 0.005 inch.

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3.2.9 Identification. Identification markings and records shall be such as to ensure positive identification of all plates, including test plates for first article test samples and specimens with the lot and corresponding heat from which they were produced. The key to identification symbols shall be furnished to the inspector prior to submittal for inspection and test.

3.2.10 Information required. A statement showing the ingot analysis of each melt and complete details of the heat treatment of each lot shall be furnished for the files of the procuring activity. All elements of the chemical composition shown in table I plus any additives or hardening agents, shall be shown in the statement. Unless otherwise specified, this information shall be forwarded on an approved form (Check List for Armor Data) along with the ballistic test plates to the proving ground making the ballistic test.

3.2.11 Workmanship.

3.2.11.1 Surface imperfections. All plates shall be free from scale, pitting, tears, and other surface imperfections which are of such a nature as to affect the fabricability or serviceability of the armor.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection.
b. Quality conformance inspection (acceptance).

4.2.1 First article inspection. The examination and tests specified in 4.5 and 4.6.1 through 4.6.4 are classified as first article inspection and shall serve as a basis for approval of the manufacturing process to be used.

4.2.1.1 Responsibility for inspection. First article examinations and tests, including ballistic tests, shall be performed at testing facilities designated by the procuring activity (see 6.3).

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4.2.1.2 Manufacturing methods. The first article samples and test plates (see 4.4.1) shall be manufactured by the processes proposed for use on production armor.

4.2.2 Quality conformance (acceptance) inspection. The acceptance examination under 4.5 and the tests under 4.6 shall serve as a basis for the acceptance of individual production lots.

4.3 Lot. A lot shall consist of all plates of the same melt of steel, of the same thickness, having the same treatment, and heat treated in the same facility.

4.4 Sampling.

4.4.1 First article samples. The number and sizes of test plates and samples for first article testing and examination shall be in accordance with table VII (see 3.2.9).

4.4.2 Production acceptance samples.

4.4.2.1 For chemical analysis. At least one sample shall be selected from each lot in accordance with method 111 or 112 of Fed. Test Method Std. No. 151.

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Table VII. Test plates and samples for first article testing and examination

Thickness of plates and samples[1/]	No. of ballistic test plates	No. of impact[2/] test samples per plate	Minimum size[3/] of ballistic test plates	To produce thickness range
Inches			Inches	Inches
3/8	3	1	18 x 36	0.188 to less than 0.50
5/8	3	1	18 x 36	0.500 to less than 0.75
1	3	1	18 x 36	0.750 to less than 1.25
1-1/2	3	1	18 x 36	0.250 to less than 2.000
2	3	1	18 x 36	2.000 to 3.000 incl.

1/ At the option of manufacturer, the exact thickness to be rolled within any range may be substituted for the specified thickness of that range.

2/ Impact samples shall be taken from test plates in accordance with 4.4.2.3.

3/ Unless otherwise specified, the average thickness of each test plate shall be determined by taking the average of 4 thickness measurements. One measurement shall be taken along each edge at a distance of at least 1 inch from the edge. Thickness measurements for plates up to and including 5/8 inch shall be read to the nearest 0.001 inch. Plates over 5/8 inch shall be read to the nearest 0.01 inch. If a producer submits plates outside the specified plate tolerances (see tables II and III) the procuring activity shall be notified.

4.4.2.2 For hardness tests. Each plate in each lot shall be tested in at least two places on each face for Brinell hardness.

4.4.2.3 For Charpy-V-notch impact tests. At least one sample shall be taken from a plate representing each lot for Charpy V-notch impact tests. The sample shall be the same thickness as the plate they represent and large enough to obtain at least two specimens from each sample in accordance with 4.6.3.

4.4.2.4 For ballistic acceptance samples. Unless otherwise specified in the contract or order, one test plate shall be submitted from each lot. The minimum plate size shall be as shown in table VIII.

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Table VIII. Minimum plate size for acceptance ballistic samples

Armor thickness (see table VII, footnote 3)	Plate size
Inches	Inches
0.188 to 1.000 incl.	12 x 36
Over 1 thru 3.000	18 x 36

4.5 Examination.

4.5.1 Visual. All steel plate shall be subject to visual inspection for compliance with the requirements for surface conditions (see 3.2.11.1) and identification marking (see 3.2.9).

4.5.2 Dimensional. All steel plate shall be subject to inspection for compliance with dimensional and tolerance requirements (see 3.2.7 and table II or III as applicable). The thickness of all plates shall be determined by measurements made one foot on centers over the entire surface.

4.5 3 Preparation for delivery. Examination shall be made to determine compliance with the requirements for preparation for delivery (see section 5).

4.6 Tests.

4.6.1 Chemical analysis. Samples for chemical analysis shall be prepared and tested in accordance with method 111 or 112 of Fed. Test Method Std. No. 151. In case of dispute, the analysis by method 111 shall be the basis for acceptance or rejection.

4.6.2 Hardness tests.

4.6.2.1 Brinell hardness test. Each plate in each lot including first article ballistic test plates shall be prepared and tested in accordance with ASTM E 10, using a 10 mm tungsten carbide ball and a 3,000 kilogram load. After removing surface scale and decarburization, the Brinell hardness testing (HB) shall be conducted on at least two locations on each face. Hardness tests may be made on the surface of pieces cut from the plate after heat treatment.

4.6.2.2 Cross-sectional hardness test. Each sample 1/2 inch in thickness through 3 inches in thickness from which impact test specimens are taken, shall have a cross-section hardness test conducted on it with at least two HB readings taken midway between the surfaces of the plate.

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4.6.3 Charpy V-notch impact test. At least two Charpy V-notch impact specimens shall be taken from each sample and prepared and tested in accordance with ASTM E 23. Specimens shall be taken in the TL orientation from a location midway between the top and bottom surfaces of the plate and at least 4 inches or 2T, whichever is less, from any quenched edge. When the amount of material available is insufficient to obtain standard specimens, the largest attainable subsize Charpy V-notch impact specimens shown in figure 7 of ASTM E 23 shall be used, and the readings converted to standard size values.

4.6.4 Ballistic test. Ballistic testing of the armor plate shall be conducted at a Government test facility specified in the contract or order to determine compliance with the requirements of 3.2.6.

4.6.5 Decarburization. Decarburization shall be determined in accordance with ASTM E 18 using the Rockwell superficial scale (15N or 30N) on hardened but untempered specimens. Each hardness determination shall be the average of three readings for the same depth below the as-heat-treated surface. The depth of decarburization is defined as the perpendicular distance from the surface to the nondecarburized zone below which there is no increase in hardness.

4.7 Reinspection.

4.7.1 First article test plates. No re-examination or retesting of originally submitted test plates that failed the first article tests shall be permitted. To achieve first article approval, the supplier shall submit three new test plates each of which shall pass the applicable examination and tests specified herein.

4.7.2 Production acceptance armor.

4.7.2.1 Examination. Acceptance armor that failed the original tests shall be reinspected for visual and dimensional examination provided that the supplier has removed all nonconforming plates.

4.7.2.2 Tests.

4.7.2.2.1 Hardness. Plates that failed because of low- or high-hardness may be retested without reheat treatment at the option of the contractor. Under such conditions, twice the original number of hardness readings shall be taken and the results of all readings taken on the plate shall be averaged to determine compliance with the requirements of table IV or V. Plates that failed because of high hardness may be retempered, in which case only those readings taken after retempering shall be used to determine acceptability. Plates that failed because of low hardness may be requenched and retempered, in which case testing shall be conducted in accordance with the requirements for original testing.

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4.7.2.2.2 First article and production impact resistance. A first article sample or production lot that failed to meet the impact resistance requirements may be retested without reheat treatment by taking not less than six additional specimens from the steel under consideration. The additional specimens shall be taken from the same location as the original tests. If the average impact resistance of all tests from the sample meets the requirements of table V for the appropriate hardness range, the lot shall be accepted, otherwise it shall be subject to rejection.

4.7.2.2.3 Ballistic properties.

4.7.2.2.3.1 Plates 1/4 thru 3 inches in thickness. A lot subject to rejection for failure to meet the requirements of 3.2.6, may be retested using the same heat treatment at the option of the contractor. Under such conditions, two additional plates shall be submitted from the lot under test. Failure of either of these plates shall be cause for rejection of the lot represented and no further testing shall be permitted on the material using the same heat treatment.

4.7.3 Resubmission.

4.7.3.1 After reheat treatment. A first article sample or production lot or portion of the lot that has failed to meet the applicable requirements specified herein may be reheat treated by quenching and tempering and resubmitted for test in accordance with the requirements for original testing. Failure of samples, ballistic test plates, or both, after reheat treatment, shall be cause for rejection of the lot they represent. However, production plates that failed to meet only the impact resistance requirements may be accepted individually on the basis of tests made on each plate.

4.7.3.1.1 Production plates that are retempered to meet only the hardness requirements shall not be subject to retest for impact resistance. Production plates requenched and retempered to meet the hardness requirements shall be tested for impact resistance and hardness as a new lot.

4.8 Reduced testing. At the discretion of the procuring activity, the amount of testing may be reduced provided the results on consecutive lots indicate that a satisfactory uniform product meeting the testing requirements is being produced. Reduced testing shall be in accordance with a system previously approved or established by the procuring activity involved.

4.9 Rejection and retest. Unless otherwise specified in the contract or order, rejection and retest or resubmittal of rejected lots shall be in accordance with the general section of Fed. Test Method Std. No. 151. Failure of the first article test specimens to meet the specified requirements shall be cause for rejection of the manufacturing process. Failure of the test specimens to meet the acceptance requirements of this specification shall be cause for rejection of the lot represented.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Unless otherwise specified in the contract or order, preservation and packaging shall be level C (see 6.2).

5.1.1 Level C. Cleaning, drying, preservation, and packaging shall be in accordance with manufacturer's commercial practice.

5.2 Packing. Unless otherwise specified in the contract or order, packing shall be level C (see 6.2).

5.2.1 Level C. All plates shall be packed in such a manner as to ensure acceptance and safe delivery by the carrier for the mode of transportation employed.

5.3 Marking. Plates (green and heat treated) shall be indelibly and legibly marked with symbols that will definitely identify the plates with inspection and test reports and with shipping documents in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The armor specified herein is intended for use on combat vehicles, but may also be applicable to armor for emplacements, shields, pill boxes, testing, and the like.

6.1.1 Class 1. Class-1 wrought steel armor is intended for use in those areas where maximum resistance to penetration by armor piercing types of ammunition is required.

6.1.2 Class 2. Class-2 wrought steel armor is intended for use in those areas where maximum resistance to failure under conditions of high rates of shock loading is required and where resistance to penetration by armor-piercing ammunition is of secondary importance. It is for protection against anti-tank land mines, hand grenades, bursting shell, and other blast-producing weapons.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Class of armor.
c. When first article samples are to be made available (see 3.1).
d. Dimensions (see 3.2.7.1).
e. When tighter flatness tolerances are required (see 3.2.7.3).
f. Name of inspection agency when inspection shall be performed by other than the contractor (see 4.1).
g. Testing facility responsible for first article testing (see 4.2.1.1).
h. When a different method of taking thickness measurements is to be used (see table VII, footnote 3.
i. When ballistic acceptance samples should be other than in 4.4.2.4.
j. The reduced testing plan when applicable (see 4.8).
k. When rejection and retest differs from 4.9.
l. Preparation for delivery requirements if necessary (see section 5).

6.3 Disposition of ballistic test plates.

6.3.1 First article test plates. Upon request of the applicant giving shipping instructions, first article plates, after ballistic testing at a Government facility, will be returned "as is" to the applicant, at his expense, unless destroyed in testing.

6.3.2 Acceptance test plates. Acceptance test plates that comply with the requirements of this specification are considered as part of the lot of steel they represent and ownership of them passes to the Government with the acceptance of that lot. Acceptance test plates that fail to comply with the requirements of this specification are considered as part of the lot they represent and remain the property of the producer just as does the rejectable lot they represent. Such failed plates will be disposed of by the testing agency unless the producer requests their return at the time test results are reported.

6.4 Definitions.

6.4.1 Fair impact. An impact resulting from the striking of the test plate, by a projectile in normal flight (no yawing or tumbling) and separated from another impact or the edge of the plate, hole, crack or spalled area by an undisturbed area of at least three calibers for plates up to and including 1.125 inches thick and an undisturbed area of at least two calibers for plates greater than 1.125 inches thick.

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6.4.2 Witness plate. A witness plate is a .014 inch thick sheet of 5052 H36 aluminum alloy (or a .020 inch of 2024-T3 aluminum alloy) placed parallel to, and 6 inches behind the test plate or other ballistic sample.

6.4.3 Protection complete penetration, CP(P). A protection complete penetration is obtained when the projectile, a projectile fragment, or a fragment of the ballistic test plate is thrown beyond the rear of the plate with sufficient energy to perforate a 0.014 inch 5052 H36 or a 0.020 inch 2024-T3 aluminum sheet placed parallel to and 6 inches ± 1/2 inch behind it.

6.4.4 Partial penetration, PP(P). Any fair impact that is not a complete penetration shall be considered a partial penetration.

6.4.5 Protection ballistic limit, BL(P). For caliber .30 AP, M2 and caliber .50 AP, M2 penetration tests, the BL(P) shall consist of an equal number of fair impact complete and partial penetration velocities attained by the up-and-down firing method. All BL(P)'s shall be computed using the highest partial penetration velocities and the lowest complete penetration velocities. Firing shall continue until either a 4 round BL(P) having a maximum velocity spread of 60 fps or a 6 round BL(P) having a maximum velocity spread of 90 fps has been attained, whichever comes first in the normal sequence of firing. In the event that the zone of mixed results (difference between the high partial penetration velocity and the low complete penetration velocity, the PP(P) velocity being higher than the low CP(P) velocity) exceeds 90 fps, the firing data will be compared with the specification minimum ballistic requirements. If the lowest complete penetration velocity is equal to or above the minimum specified ballistic limit velocity for the plate thickness, the ballistic limit will be computed on the basis of 4 or 6 rounds using the smallest possible velocity spread. If the lowest complete penetration velocity is below the minimum allowable ballistic limit velocity then testing will continue until a ten round ballistic limit has been attained using the smallest possible velocity spread. Ten round ballistic limits will be reported as such on the armor data forms TAC 3983.

For 37 mm AP, M74 projectiles, the protection ballistic limit is the average of four fair protection criteria impact velocities comprising the first two lowest velocities in the firing order resulting in complete penetrations and the first two highest velocities in the firing order resulting in partial penetrations which meet the condition that the velocity spread for the four rounds shall not exceed 90 fps.

6.5 Thickness. The symbol "T" is used throughout this specification to indicate the nominal thickness of the plate under consideration.

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6.6 Charpy V-notch impact specimens used for comparison tests should be obtained from the local DSA (DCASR) office or, in cases where DCASR does not have responsibility over a particular contract, from the U. S. Army Materials and Mechanics Research Center, Watertown, MA 02172.

Custodian:
Army - MR

Preparing activity:
Army - MR

Review activity:
Army - AT

Project No. 9515-A018

User activity:
Army - MI

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel (either government or contractor) involved in the use of the specification in the procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification with a view to insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this sheet will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION MIL-A-46173(MR), Armor, Steel, Plate, Wrought, (ESR) (3/16 Through 3 inches, Inclusive)

ORGANIZATION | CITY AND STATE

CONTRACT NO | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (*Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.*)

SUBMITTED BY (*Printed or typed name and activity*) | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863 WHICH IS OBSOLETE

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